Similarity-Based Interference in L2 Production of English Object Relative Clauses

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Introduction: Research on object relative clause (ORC) processing shows that *similarity-based interference (SBI)* plays a key role in processing difficulty[1-3]. When two noun phrases share the same animacy configuration, production become more difficult due to interference during memory retrieval. Elicitation studies show that L1 speakers of various languages favor passives (e.g., *the baby carried by the woman*) over actives when both nouns are animate, reflecting SBI[6,8], as passive structures are thought to mitigate this interference by linearly separating the two similar nouns[5,9]. In L2 processing research, some authors proposed that SBI may be more disruptive for L2 speakers due to shallow processing[4] or reduced memory capacity[5]. However, Cunnings and Fujita (2023) found that L1 and L2 speakers showed similar interference patterns in ORC comprehension. Less is known about SBI in L2 ORC production. While Izumi (2003) found that L2 learners show greater difficulty producing active ORCs than L1 speakers, he did not investigate animacy effects. To fill this gap, the current study examines animacy-driven interference patterns in L1 and L2 production of English ORCs. We recruited L1-Spanish and L1-Mandarin speakers to perform an elicitation task in L2 English, with L1-English speakers as baseline. We used four animacy conditions (Table 1), including a novel **II** condition (both nouns inanimate). ***Hypothesis1*** (L1-L2 difference): Following Izumi (2003), L2 speakers may favor passives more strongly than L1 speakers due to the high processing demands of active ORCs. ***H2*** (Typological transfer): Spanish and English share postnominal RC structure, while Mandarin has prenominal RCs; thus, we expect L1-Spanish speakers to pattern more similarly to L1-English than L1-Mandarin speakers. ***H3*** (Animacy sensitivity): Given cross-linguistic evidence for animacy effects in RC production of Spanish[8] and Mandarin[6], animacy effects should emerge in L2 production, but whether L2 sensitivity is comparable to L1 remains an open question, with competing accounts positing parallel[3] versus distinct[4] patterns.

Method: *Participants (Table.2):* 116 adults: L1-English (N=38, baseline), L1-Spanish (N=39), and L1-Mandarin (N=39). L2 participants completed LexTALE proficiency assessments and a Backward Digit Span Task (WM). *Picture-Based Elicitation*: 20 oral responses (plus 40 fillers) were elicited per participant across four animacy conditions using pictures and question prompts. Structure choice (passive vs. active) was measured.

Results: L1-English speakers showed clear animacy-driven variation, with the fewest passives in IA and the most in AA, confirming influence of both animacy congruency and head noun animacy (Fig.1). L1-Spanish speakers patterned similarly to L1-English across all conditions (Fig.1). In contrast, L1-Mandarin speakers showed near-ceiling passive use (~95%) with no reliable head noun or congruency effects (Fig.2). *Working Memory (Fig.3):* Although WM did not interact with animacy, a main effect of WM on structure choice emerged: L1-Spanish speakers with higher WM produced fewer passives (i.e., more actives) (β = -1.01, p = .047), mirroring the L1-English baseline pattern. L1-Mandarin speakers showed no WM effect (p = .91)., likely due to ceiling-level passive use. *Language Proficiency (Fig.4):* LexTALE did not modulate the animacy effect for either group. However, a main effect of proficiency emerged for L1-Spanish speakers: higher English proficiency predicted fewer passives (β = -1.20, p = .019). No proficiency effect was found for L1-Mandarin speakers (p = .41).

Discussion: Our findings reveal that L1-Spanish speakers showed animacy effects parallel to L1-English, consistent with Cunnings & Fujita's (2023) findings for similar patterns of SBI in L1 and L2 comprehension. In contrast, L1-Mandarin speakers showed a strong passive preference regardless of animacy. While this could reflect heightened interference, we suggest that L1 influence from Mandarin is more parsimonious. The WM main effect for L1-Spanish speakers (higher WM→more actives) suggests that active ORCs are generally more demanding to produce than passives, while animacy sensitivity itself may operate independently of WM. Similarly, the proficiency effect for L1-Spanish indicates that increased proficiency level generally facilitates production of active ORCs. In conclusion, we argue that L2 speakers are *not* inherently more susceptible to SBI in production than L1 speakers are, extending Cunnings & Fujita's (2023) approach from comprehension to production.

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		Head Noun	
		Animate (A)	Inanimate (I)
Agent Noun	Animate (A)	**AA** ACTIVE: *the baby that the woman lifts* PASSIVE: *the baby that's lifted by the woman*	**IA** ACTIVE: *the rock that the athlete lifts* PASSIVE: *the rock that's lifted by the athlete*
	Inanimate (I)	**AI** ACTIVE: *the tiger that the crane lifts* PASSIVE: *the tiger that's lifted by the crane*	**II** ACTIVE: *the barrel that the helicopter lifts* PASSIVE: *the barrel that's lifted by the helicopter*

Table 1. Animacy conditions for target productions

	L1-Spanish (N = 39)	L1-Mandarin (N = 39)
Age	30.3 (8.2) [19–59]	26.1 (5.1) [11–38]
Age of Acquisition	7.2 (5.5) [5–21]	7.4 (3.2) [3–19]
Years in English environment	9.1 (8.7) [0–21]	3.8 (3.4) [0–13]
LexTALE score	82.7 (12.2) [56.2–98.8]	71.8 (13.4) [48.8–96.2]
BDS Span	6.5 (2.3) [2–10]	8.3 (2.0) [1–10]
BDS Score	25.2 (9.7) [2–43]	33.4 (9.6) [2–45]
Gender (F/M/NB)	23/15/1	21/18/0

Note. Values are M (SD) [range]. BDS = Backward Digit Span; F = female; M = male; NB = non-binary.

Table 2. L2 participants profile

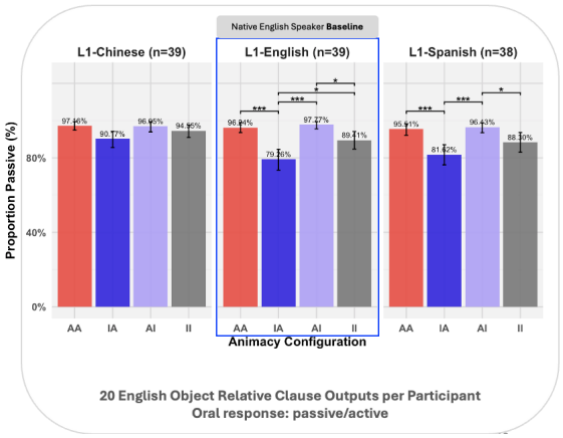

Figure 1. Passive ORC Proportion by Animacy

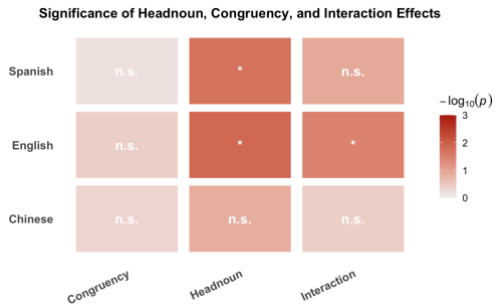

Figure 2. Reaction Time by Animacy

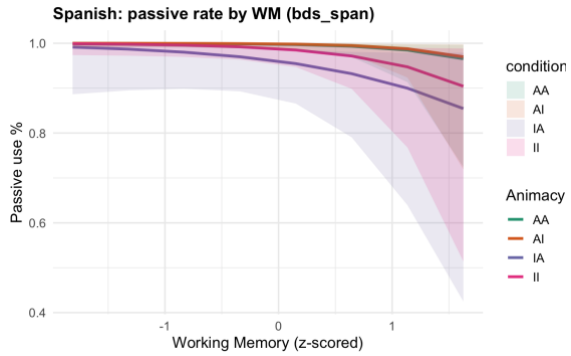

Figure 3. Passive usage rate by working memory span among L1-Spanish learners of English

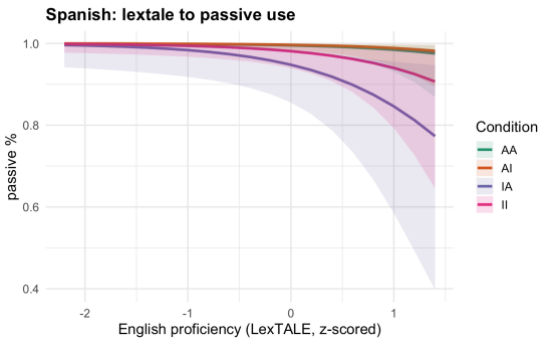

Figure 4. Passive usage rate by English proficiency level among L1-Spanish learners of English

Selected References:

1. Gordon, P. C., Hendrick, R., & Johnson, M. (2001). Memory interference during language processing. *Journal of Experimental Psychology: LMC*, *27*(6), 1411-1423.
2. Cunnings, I. (2017). Parsing and working memory in bilingual sentence processing. *Bilingualism: Language and Cognition*, *20*(4), 659-678.
3. Cunnings, I., & Fujita, H. (2023). Similarity-based interference and relative clauses in second language processing. *Second Language Research*, *39*(2), 539-563.
4. Clahsen, H., & Felser, C. (2006). Grammatical processing in language learners. *Applied Psycholinguistics*, *27*(1), 3-42.
5. Gennari, S. P., Mirković, J., & MacDonald, M. C. (2012). Animacy and competition in relative clause production. *Cognitive Psychology*, *65*(2), 141-176.
6. Hsiao, Y., & MacDonald, M. C. (2016). Production predicts comprehension: Animacy effects in Mandarin relative clause processing. *Journal of Memory and Language*, *89*, 87-109.
7. Izumi, S. (2003). Processing difficulty in comprehension and production of relative clauses by learners of English as a second language. *Language learning*, *53*(2), 285-323.
8. McDonald, J. L. (2006). Beyond the critical period: Processing-based explanations for poor grammaticality judgment performance by late second language learners. *Journal of Memory and Language*, *55*(3), 381-401.
9. Rodrigo, L., Igoa, J. M., & Sakai, H. (2018). The interplay of relational and non-relational processes in sentence production. *Frontiers in Psychology*, *9*, 325103.